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# Federated Node (FN): an open-source, GA4GH TES-based service for building federated research networks

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## Summary

The Federated Node is an open-source software component for running federated analytics. It is based on existing open standards and is designed to be easy to deploy, manage and integrate with existing infrastructure within data controller's environment for a wide variety of data types and use cases. It has been developed for the PHEMS consortium and is already deployed at multiple partner sites, forming the basis of their federated data sharing network.

## Statement of Need

Collaborative biomedical and clinical research increasingly depends on access to diverse, curated real-world data. However, despite progress in open science and digital health infrastructure, such collaboration remains fragmented. Data are often siloed across institutions, jurisdictions, and governance frameworks, particularly in domains such as rare disease and cancer research where sample sizes are inherently small and geographically dispersed. Harmonising and analysing these data collectively are essential for scientific reproducibility and translational impact (Legido-Quigley et al., 2025).

Efforts to promote open and FAIR (Findable, Accessible, Interoperable, Reusable) data practices have revealed the natural tension between data accessibility and usability, and privacy protection. The introduction of robust privacy legislation such as the EU's General Data Protection Regulation (GDPR) and equivalent frameworks worldwide has been vital in safeguarding individual rights. Yet, the resulting regulatory and technical fragmentation has made it increasingly difficult for researchers to move, share, or co-analyse data across national borders e.g. (Mourby et al., 2019). The European Health Data Space (EHDS) and similar policy initiatives recognise that secure federated approaches, where data remain within their source environments but can be analysed collectively, are essential to balance privacy with scientific utility.

Federation has therefore emerged as a practical and ethical mechanism for enabling cross-border research. Rather than copying or aggregating sensitive datasets into central repositories, federated networks allow analytical code to be executed remotely under the control of data custodians, ensuring data never leave institutional boundaries. This paradigm directly addresses legal and ethical constraints on cross-border data movement while maintaining auditability and governance alignment with local policies (Eradat Oskoui et al., 2025). The success of nationwide federated EHR networks in routine emergency-care research (Bienzeisler et al., 2025) demonstrates that such infrastructures are now technically feasible and not merely conceptual.

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## State of the field

While the technical foundations for federated analytics and learning are in place, translating them into practice remains challenging. Even comprehensive frameworks such as the GA4GH Task Execution Service and the ICODA Common API cannot, by themselves, overcome the deep heterogeneity that exists in real-world healthcare systems. Recent reviews of federated learning in medicine emphasise that most studies still fail to reach clinical utility, citing issues such as non-identical data distributions, methodological bias, heavy communication overheads, and incomplete governance alignment (Joshi et al., 2022; Li et al., 2025). A recent systematic review found that of more than 22,000 papers screened, fewer than 6 per cent involved genuine real-world deployments, underscoring persistent barriers to clinical translation (Teo et al., 2024). Large-scale national programmes such as Australian Genomics further highlight the operational and governance coordination required even before federated analytics are introduced (Stark et al., 2023). These findings collectively illustrate that while federation is technically feasible, its successful implementation demands standards-aligned infrastructure that reduces operational friction and lowers the barrier for adoption.

Despite these advances, establishing and maintaining interoperable federated networks remains complex. Implementations often require substantial local customisation to integrate with existing authentication, container orchestration, and registry systems. There is therefore a pressing need for an open and reproducible reference implementation that lowers the barrier for organisations to participate in federated research while building on the GA4GH Task Execution Service (TES) specification.

The Federated Node (FN) addresses this need by operationalising the ICODA Common API into a deployable, open-source package built from widely adopted components (Keycloak, Traefik, PostgreSQL, Kubernetes). FN allows institutions to host secure, standards-based endpoints capable of executing authorised analytical tasks against local datasets, thereby enabling scalable, privacy-preserving, cross-institutional collaboration aligned with international standards.

By operationalizing existing specifications rather than creating proprietary approaches, the FN lowers adoption barriers while supporting interoperability, as illustrated by its use in the PHEMS network described below.

## Research Impact Statement

The FN was developed for, and is used within, the PHEMS (Pediatric Hospitals as European drivers for multi-party computation and synthetic data generation) consortium, a collaboration between four European children's hospitals and industry partners to build a federated network for paediatric and rare disease research. Each participating hospital deploys an FN in its own environment, while analytical code is managed in a shared GitHub organisation; approved changes are pulled into each hospital's FN via ArgoCD, so no inbound connections to hospital networks are required (Stiven, 2025).

The first test of the core network, in March 2025, ran the PHEMS clinical benchmarking use case on data mapped to the OMOP Common Data Model, using FNs deployed at Great Ormond Street Hospital and Helsinki University Hospital, with results returned either to GitHub or directly to an analysis workspace (Stiven, 2025). Onboarding of the four founding hospitals followed (Stiven, 2026a), and the network now provides a production-ready federated benchmarking capability that allows members to compare clinical outcomes (Stiven, 2026b). The network is intended to form the basis of the Paediatric Health Data Space, presented publicly in 2026 (i4KIDS, 2026; Stiven, 2026a).

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## Software Design

The International Covid-19 Data Alliance (ICODA) developed the Federated Data Sharing Common API or "Common API" as an open standard for a federated data sharing API, as described elsewhere. The Common API is a constrained implementation of the Global Alliance for Genomics and Health Task Execution Service, containing components for meta-data browsing, remote data selection and federated computation.

The Federated Node (FN) builds upon this open standard, and provides a practical, working implementation of the standard that can be easily deployed and operated within federated research networks. The FN packages an implementation of the Common API with other open-source components including:

- The Common API
- Keycloak
- Traefik, via the Kubernetes Gateway API

The Common API specifies a set of endpoints that provide a framework for organisations that wish to collaborate on federated data sharing and analysis. It provides the structure of the Federated Node API.

The FN implements the Common API rather than the full TES specification. Its task endpoints follow the TES model (`POST /tasks` to submit a task, `GET /tasks` and `GET /tasks/{id}` to list and inspect tasks, and `POST /tasks/{id}/cancel` to cancel one), but differ from TES in some details, such as the form of the cancel path. The FN also adds endpoints that TES does not define, for dataset metadata, container registries, audit logs and result review.

Keycloak is used for token and user management. Incoming requests are routed through Kubernetes Gateway API resources (a Gateway and HTTPRoutes), with Traefik as the gateway controller; the Helm chart can optionally install Traefik itself. Releases before 1.8.0 used nginx-ingress instead.

Federated Node deployments use widely adopted open-source technologies. FNs are deployed to a Kubernetes cluster using a Helm chart and require a PostgreSQL database, which holds Keycloak's identity data and the FN's own records of datasets, registries, containers, tasks and audit events. Each Federated Node is also associated with one or more configured container registries (for example Azure Container Registry, GitHub Container Registry or other Docker-compatible registries), which hold the task images that can be run against the data. Task results are written to configured persistent storage (local, NFS, Azure Files or AWS EFS volumes at the reviewed release).

Figure 1 below describes how a federated task is processed when initiated by an authenticated user:

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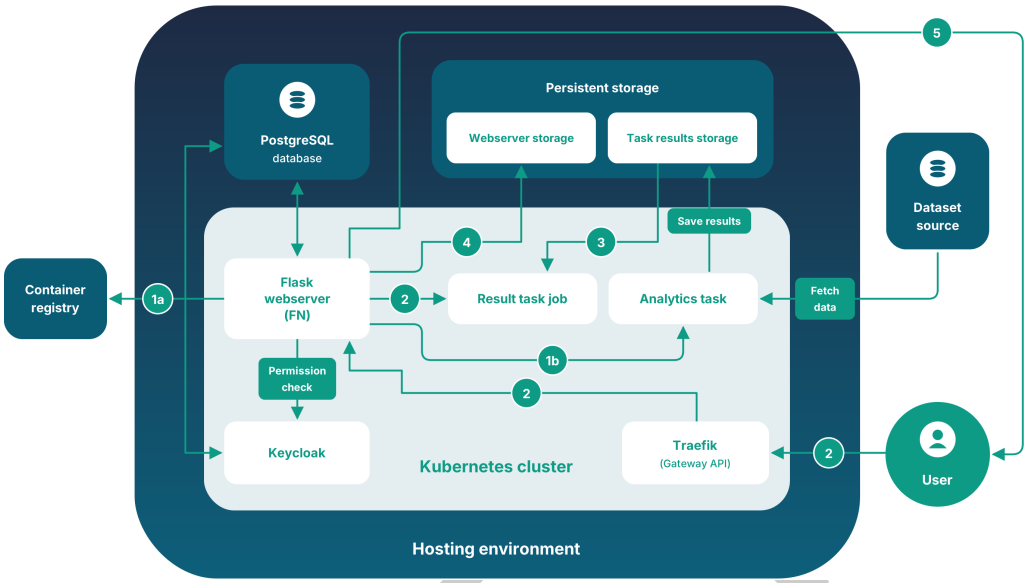



**Figure 1:** Federated task processing.

- *1a* Before creating the task pod, the FN checks that the requested container image is available in one of the container registries configured for the FN.
- *1b* The task pod is created, and the results are saved in the condfigured persisten storage.
- *2* On calls to the `/tasks/{id}/results` endpoint, if the task pod is on completed status, a job is created.
- *3* The job's pod mounts both the task's and the webserver's persistent storage. It fetches the task's results folder and zips it.
- *4* The webserver reads the zip contents from the live job pod and saves it in its own presistent storage.
- *5* The resulting archive is returned to the end user

This architecture gives data owners several layers of control over what is run against their data and what leaves their environment. Task images must be registered with the FN and held in one of its configured container registries; users authenticate through Keycloak and need the appropriate role and dataset permission to submit tasks; non-admin users can only access their own tasks; and task-related API calls are recorded in an audit log. Data owners can additionally enable result review, in which case results are released to users only after an administrator approves them. These controls operate alongside the hosting institution's own Kubernetes RBAC, network and registry access policies.

## Experimental AI extension

An experimental fork, Federated Node AI, explores adding an `/ask` endpoint that sends a user's prompt, together with data retrieved locally by the FN, to a small language model hosted in the same environment as the data. This work is at an early stage, is not part of the software described in this paper, and significant security and reliability questions would need to be resolved before such an approach could be used with sensitive data.

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## PHEMS

The following members of the PHEMS ("Pediatric Hospitals as European drivers for multi-party computation and synthetic data generation capabilities across clinical specialities and data types") consortium contributed substantially to this work from their institutions over the course of the project, including the use cases, requirements and validation for the FN. Because of the size of the consortium, and at its request, they are credited collectively through the group author PHEMS rather than listed individually in the author list:

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## Author contributions

R.C. and R.S. designed and developed the software. S.R., R.B. and D.S provided oversight of the project. PHEMS contributed use cases that informed requirements and validation. All authors contributed to the writing and/or review of the manuscript.

## AI usage disclosure

No generative AI coding assistants were used in writing the Federated Node code up to version 1.9.0, the version submitted for review. Some later changes included fixes suggested by GitHub Copilot Autofix, and Claude (Anthropic) was used to draft design documents on development branches. Generative AI tools may also have been used to assist with parts of the project documentation. All such changes were reviewed by the development team before being merged.

The original manuscript was written by the authors without AI assistance. During revision in response to peer review, Claude Opus (Anthropic) was used to cross-check the manuscript against the source code and reviewer comments, identify inaccurate statements, locate public supporting sources, and draft revised wording for several sections, including the Research Impact Statement and Software Design. The authors reviewed, edited and verified all suggested text against the code and the cited sources, made all decisions about content, and take full responsibility for the paper.

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